Physical Properties of Alcohol Based Deep Eutectic Solvents

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ABSTRACT

The physical properties, viscosity, conductivity, surface tension and density were measured for a series of alcohol based Deep Eutectic Solvents (DES) as a function of choline chloride (ChCl) concentration. The point at which the system becomes a saturated salt solution in the diol systems is at 20 mol% ChCl, because at higher concentrations the conductivity decreases despite the increase in number of charge carrying species. The addition of ChCl to glycerol results in a decrease in the viscosity of the system by interrupting the intermolecular hydrogen bonding of the glycerol molecules, whereas the addition of ChCl to the diols results in an increase in viscosity. The mono-alcohol phenol has been shown to form a DES in a ratio of 4:1 phenol: ChCl. Stable DES's have also been formed using sugars, D-(-) fructose, D-glucose, xylitol and *meso*-erythritol.

Determination of solvent-solute interactions is key to the understanding of solvent properties in liquids therefore the Kamlet-Taft parameters α, β, π* and $E_T(30)$ parameter were determined for a series of alcohol based DES using three solvatochromic dyes, 4-nitroaniline, N,N-dimethyl-4-nitroaniline and Reichardt's dye $E_T(30)$. It is shown that they are similar to $RNH_3^+X^-$, $R_2NH_2^+X^-$ and imidazolium ionic liquids. The effect of ChCl on the Kamlet-Taft parameters is not uniform across the systems and is dependant upon the hydrogen bond donor.

The solventless esterification of glycerol and lauric was carried out using ChCl with the intention of using its hydrogen bonding interaction with glycerol to behave a potential site inhibitor as to selectively form mono- or di-ester products. The work has shown that the addition of ChCl to the system, initially slows the rate of the mono-ester formation, but actually encourages the reaction to progress further towards the di-ester product, when compared to the pure glycerol system encouraging only mono-ester formation.

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APPENDIX I

CHAPTER 1

Introduction

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1 Introduction

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1. Introduction

Ionic liquids are salts that have a very low melting temperature.[1] By definition they are molten salts that are required to melt below 100 °C.[1,2] Ambient melting requires that the melting point of the ionic liquid occur at a temperature that is not very far from the glass temperature.[3] The majority of ionic liquids have melting points in the range of;[4]

$T_g – 1.5T_g$ **(1.1)**

Structurally, at least one of the ionic components is an organic compound, most commonly the cation.[5] The ions in ionic liquids are both large and bulky with the cationic component tending to have a very low degree of symmetry. By carefully considering these components it is possible to prepare ionic liquids that are liquid around and below room temperature.[1,2,5]

In recent years, a lot of interest has been generated in ionic liquids for their potential use as a solvent media. This is attributed to the unique combination of properties they possess. Firstly, they are good solvents for a wide range of both organic and inorganic materials.[1,2,5] As a consequence, it is possible to bring unusual combinations of reagents into the same phase.[1,5] As ionic liquids are composed mainly of poorly coordinating ions and so as a result have the potential to be highly polar[1,2,5] yet non-coordinating solvents.[1] They are also immiscible with a large number of organic solvents, thus provide a non-aqueous, polar alternative for two-phase systems.[1] They are considered to be generally non-volatile and so have the potential to be used in high vacuum systems, although it has been reported that some ionic liquids do have significant vapour pressure.[1,2,5-8]

1.1 History of Ionic Liquids

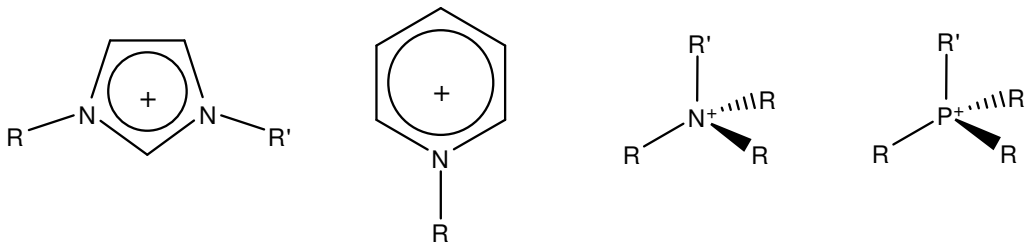
Ionic liquids, although have only gained much interest relatively recently, have actually been documented as early as 1914.[9] This ionic liquid was made from $[EtNH_3][NO_3]$, and had a melting point of 12 °C. In 1948 at the Rice Institute in Texas, the first ionic liquids containing chloroaluminate ions were developed and were used as bath solutions for electroplating aluminium.[10] By the 1980's chloroaluminate melts began to be used as non-aqueous, polar solvents for the investigation of transition metal complexes.[11-13] This work mainly began with the study of the electrochemical aspects of

transition metal complexes and was soon followed by investigations into spectroscopic and complex chemistry.[14] It was towards the end of the 1980's that ionic liquids started to be used as both a new reaction media and as catalysts. Various Friedel-Crafts reactions were carried out using acidic ionic liquids with chloroaluminate ions.[15] In addition phosphonium halides melts were used to successfully perform aromatic substitution reactions.[16] During the 1990's, weakly acidic chloroaluminate melts were used for the first time in transition metal catalysis. By dissolving nickel catalysts in the chloroaluminate melt, ionic catalyst solutions were formed and used for propene dimerization.[17] Ethylene polymerizations using the Ziegler-Natta catalyst were carried out in the weakly acidic chloroaluminate melts.[18] In 1992, the concept of ionic liquids received a substantial boost to there potential when low melting tetrafluoroborate melts were first described.[19] These systems showed a significant improvement in the stability of the solvent against hydrolysis. Tetrafluoroborate melts have a much higher tolerance versus functional groups, resulting in a much larger range of potential applications, especially for transition metal catalysis.[5] For example, the rhodium catalysed hydroformylation of olefins have been used successfully with tetrafluoroborate melts.[20]

More recent publications show that the focus of ionic liquid research has shifted toward the design of the ionic liquid itself.[21-27] Systematically, both the physical and chemical properties of the ionic liquids have been investigated.[27-30] Further applications as solvents and catalysts were then investigated as a result.[31-40]

1.2 Common Components of Ionic Liquids

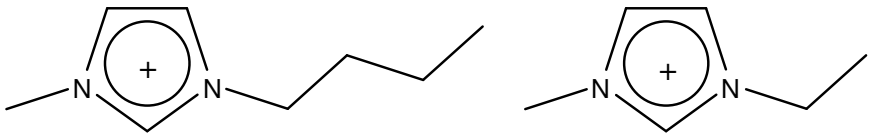
The scope for making ionic liquids is currently believed to of the order of 10^{30} compounds. These compounds are mainly made from four groups of compounds. Four common cationic groups that have been used for making ionic liquids are shown in Figure 1.1



N,N'-dialkylimidazolium **N-alkylpyridinium** **Alkyammonium** **Alkylphosphonium**

***Figure 1.1**: Common structures of cations used in ionic liquids*

In recent years, N,N'-dialkylimidazolium has become the most widely used cationic type to make ionic liquids. Two of the most widely studied N,N'-dialkylimidazolium species are 1-butyl-3-methylimidazolium (BMIM) and 1-ethyl-3-methylimidazolium (EMIM) cations (Figure 1.2).



BMIM **EMIM**

***Figure 1.2**: Structures of 1-butyl-3-methylimidazolium (BMIM) and 1-ethyl-3-methylimidazolium (EMIM)*

The anionic components commonly used in ionic liquids tend not to be as asymmetric as the cationic components, if at all. However, like their cationic counterparts they are still bulky. Common anionic components range from inorganic anions such as $[BF_4]^-$, $[PF_6]^-$, $[AlCl_3]^-$ to simple acids such as $[HNO_3]$ which form by the neutralization of the amine by the acid. Hagiwara et al. showed that fluorohydrogenate anions can be used to form room temperature liquids with N,N'-dialkylimidazolium cations.[24] The anion is in the form $(HF)_{2.3}F^-$ and is composed of a mixture of $(HF)_2F^-$ and $(HF)_3F^-$ anions. Ionic liquids containing a combination of sulfonyl and fluorine groups such as bis(trifluoromethanesulfonyl)amide $[(CF_3SO_2)_2N]^-$ has been shown to offer large depressions in the freezing points of systems.[41]

Zwitterionic liquids have generated much interest in recent times due to the component ions not being able to migrate along a potential gradient.[42-44] They show some ionic conductivity when metal salts are added and so can be used as ion conductors.

There has also been some work that has made ionic liquids by using poly-ionic species. Whilst it would appear that ionic liquids are made up of the general formula A^+B^-, it has been shown that polycationic ionic liquids which incorporate a series of polyammonium halide salts and phosphate ions, resulting in ionic liquids with the basic formula $A^{n+}B^{n-}$ can be formed.[45]

1.3 Synthesis of Ionic Liquids

The synthesis of ionic liquids usually begins with the quaternisation of an imidazole or amine, for example, to produce the cationic component.[46] The quaternisation is carried out using an alkylating agent, which is commonly a halogenoalkane. The anionic component is dependent on the form of the alkylating reagent. The alkylating agent is often a halogenoalkane. It is possible to produce salts that melt below 100 °C by this process.[1,5] Ionic liquids that can be formed by direct quaternisation for example are [BMIM]Cl and [EMIM]CF_3SO_3 etc.[5] It is however not always possible to form the desired anion directly by this process. An additional step is required. Two possible steps to generating the anionic component can be seen in Figure 1.3.

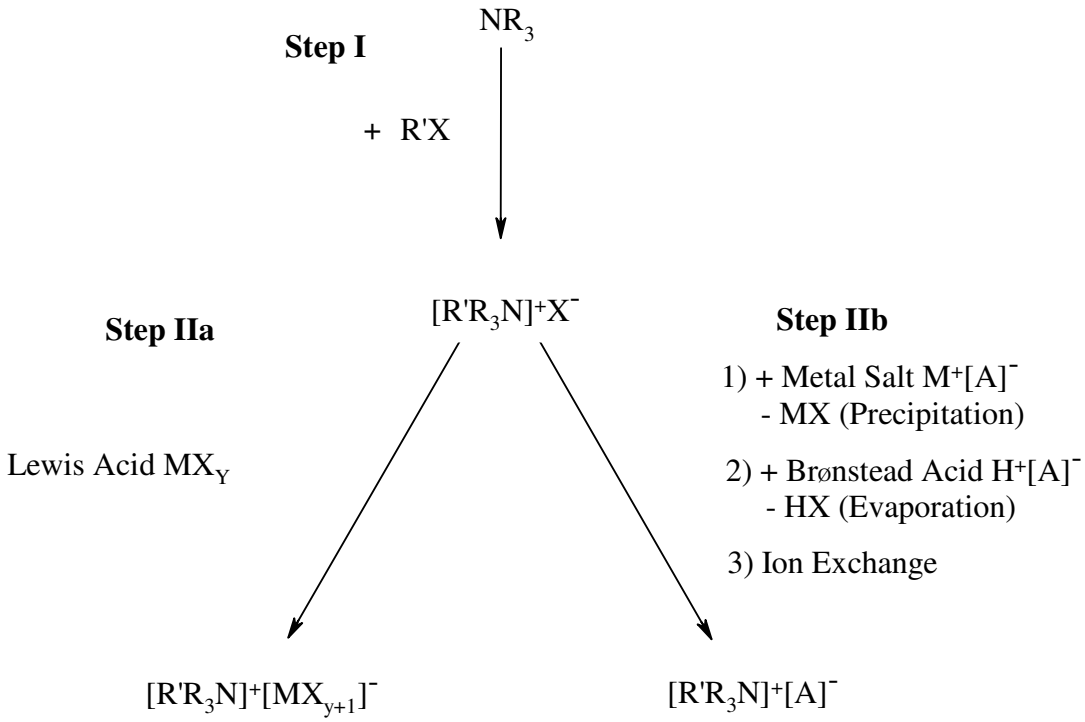


***Figure 1.3**: Synthetic pathway for the preparation of ionic liquids, examplified for an ammonium salt*[5]

Step I in Figure 1.3 describes the synthesis of an ammonium halide, $[R'R_3N]^+X^-$. From this point the ammonium halide can be treated with a Lewis acid MX_y, leading to an ionic liquid that is of the form $[R'R_3N]^+[MX_{y+1}]^-$ as is shown in Step IIa and equation (1.2) In this process several anion species are observed in equilibrium. The ratio of species present depends upon the ratio of the two components, in this case $[R'R_3N]^+X^-$ and MX_y. A good example of this is displayed by the chloroaluminate melts. As the ratio of chloride to $AlCl_3$ ratio changes, so does the anionic species. When the ratio of Cl:$AlCl_3$ the only anionic species observed is $AlCl_4^-$. When the molar fraction of $AlCl_3$ is at $x(AlCl_3)>0.5$ the chloroaluminate anions become multi-nuclear. The various species can be seen in equations (1.3) and (1.4).[5]

$$[R'R_3N]^+X^- + MX_y \rightleftharpoons [R'R_3N]^+[MX_{y+1}]^- \quad \textbf{(1.2)}$$

$$[R'R_3N]^+AlCl_4^- + AlCl_3 \rightleftharpoons [R'R_3N]^+[Al_2Cl_7]^- \quad \textbf{(1.3)}$$

$$[R'R_3N]^+Al_2Cl_7^- + AlCl_3 \rightleftharpoons [R'R_3N]^+[Al_3Cl_{10}]^- \quad \textbf{(1.4)}$$

Although, chloroaluminates are perhaps the best know ionic liquids that are formed by the use of a Lewis acid, ionic liquids containing $SnCl_2$, CuCl, BCl_4 etc. as the Lewis acid have been reported.[5]

Step IIb in Figure 1.3 also describes the production of an ionic liquid by the addition of a metal salt $M^+[A]^-$ or by displacing it with a strong acid $H^+[A]^-$. This route will only produce one anion species unlike the Lewis acid method described by Step IIa. Anion exchange methods however tend to result in impurities from residual halide species. These halide ions may attack solute materials and so it is important to remove them. Unlike conventional solvents, purification cannot occur by distillation. Purification of the ionic liquid is achieved by the use of an anion exchange resin.[5]

1.4 Physical Properties of Ionic Liquids

By careful selection of suitable cations and anions, it is possible to vary the physical properties of ionic liquids and so systems can be tailored to meet the criteria of a specific application. This has lead to them being termed 'designer solvents'.[47]

1.4.1 Melting Point

Since ionic liquids are defined as molten salts that melt below 100 °C, the melting point is a defining feature of ionic liquids. The structure and chemical composition of an ionic liquid is related to its melting point. By comparing various chloride salts, the influence of the cationic component becomes obvious. Alkali metal chlorides such as NaCl and KCl have extremely high melting points (801°C and 772 °C respectively). By replacing the cation with a suitable organic cation for example $[EMIM]^+$ or $[BMIM]^+$, the melting point decreases drastically (87 °C and 65 °C respectively).[46,48] Factors which result in lower melting ionic liquids are cations with low symmetry, weak intermolecular bonding, for example the avoidance of hydrogen bonding [49,50]and a good distribution of charge in the cation. [48-51]

The cationic component is not the only deciding factor in the melting point of the system. The anion also plays a role. By keeping the cation constant, in this case $[EMIM]^+$ it can be seen that altering the size of the anion whilst maintaining the same charge alters the melting point for example [EMIM]Cl has a melting point of 87 °C, but the melting point lowers when the Cl anion is replaced with for example NO_3 (38 °C)[46],

$AlCl_4$ (7 °C)[52], CF_3CO_2 (-14 °C).[21] As anion size is increased there is a further decrease in the melting point of the system. Also the presence of several anionic species have also been shown to reduce the melting point of the system.[52] As discussed in Section 1.3, the molar ratio of Lewis acid species added to the halide salt can alter the anionic species. This mixture of anionic species results in a decrease in the melting point of the system.

1.4.2 Vapour Pressure and Thermal Stability

It has been widely reported that ionic liquids have no measurable vapour pressure. This property means that it is possible to separate the reaction mixture by distillation effectively. There is no azeotropic effect between the ionic liquid and the product.[1,5] The thermal stability of an ionic liquid is directly related to the strength of the heteroatom-carbon and heteroatom-hydrogen bonds. Ionic liquids formed from either protonation of amines or phosphanes show significantly restricted thermal stability.[5] Many of these denature at 80 °C in vacuo.[5] Ionic liquids formed from alkylation of amines or phosphanes show higher thermal stability but do show a tendency to undergo retro-quaternization at temperatures nearer 150 °C. It has been shown that certain ionic liquid however show very high thermal statbility for example [EMIM][BF_4] has a liquid stability range of 300 °C[53] and [EMIM][$(CF_3SO_2)_2N$] has been shown to have a liquid range of 400 °C.[21]

The non-volatility so often refered to in ionic liquid literature has been shown be somewhat of an over assumption. Seddon *et al*. has shown that some ionic liquids are infact distillable under vacuum and at high temperatures.[6] It has been shown that [EMIM][$(CF_3SO_2)_2N$] (1-ethyl-3-methylimidazolium bistriflamide) and [DMIM][$(CF_3SO_2)_2N$] (1-decyl-3-methylimidazolium bistriflamide) can be distilled at a rate of 0.120 gh^{-1} and 0.070 gh^{-1}, at 300 °C and at 0.1 mbar pressure. Negligible decomposition was seen in the distillate. It has also been reported that several distillable ionic liquids have been developed based on N,N-dialkylammonium N′,N′-dialkylcarbamate melts, (dialcarbs). Dialcarbs are mixtures of CO_2 and dialkylamines.[7,8] By mixing $MeNH_2$ with CO_2 in a ratio of 1.8:1, dimethylammonium dimethyl carbamate, (DIMCARB), is formed and is a stable liquid at room temperature. During the distillation process, the ionic liquid decomposes into its constituent elements at temperatures greater than 60 °C, but can be condensed and reassociated easily, with no

decomposition of the ionic liquid. This ionic liquid has been used in both electrochemical applications and organic synthetic applications.[6-8]

1.4.3 Density

The density of an ionic liquid is related to its ionic components. By taking a series of chloroaluminate melts and only changing the cation. Wilkes *et al*. showed that there is an almost linear relationship between the length of the *N*-alkyl chain on the imidazolium cation and the density was observed.[52] The density decreased as the *N*-alkyl chain increased. As a general rule, the bulkier the cation the lower the density tends to be. Some slight structural changes allow for finer adjustments to the density. It has also been shown that the anionic effects are greater than the cationic effects. A classic example of this is to compare the [EMIM]$AlBr_4$ and [EMIM]$AlCl_4$ ionic liquids.[51] The density for the bromoaluminates is much higher than that of the chloroaluminates, ranging from approximately 0.5 gcm^{-3} – 0.8 gcm^{-3}.[54]

1.4.4 Viscosity

The strength of van der Waals interactions and tendency to form hydrogen bonds are the major factors which contribute to the viscosity of an ionic liquid.[21] Hydrogen bonding plays a large part in the viscosity of the chloroaluminate systems. High viscosity at low $AlCl_3$ molar composition due to greater hydrogen bonding between the cation and basic chloride ion.[52] At higher $AlCl_3$ molar compositions the species $AlCl_4^-$ and $Al_2Cl_7^-$ are present, resulting in a more acidic environment. The negative charge is more widely distributed through these species which results in decreasing the strength of the hydrogen bonds. The effect of van der Waals forces on viscosity is obvious in some of the hydrophobic ionic liquids containing fluorinated anions. The ionic liquid [BMIM][CF_3SO_3] has a viscosity of 90 cP at 20 °C.[21] By changing the triflate anion to [*n*- $C_4F_9SO_3$]$^-$ there is a significant increase in viscosity to 373 cP. A similar trend is observed with the use of a trifluoroacetate ion. For example [BMIM][CF_3COO] has a viscosity of 73 cP, [BMIM][*n*-C_3F_7COO] has a viscosity of 182 cP. In each case the rise in viscosity is due to an increase in van der Waals interactions for the longer chain anions. There is however an exception to this observation of stronger van der Waals interactions resulting in greater viscosity. By using the

bis(trifluoromethanesulfonyl)amide $[(CF_3SO_2)_2N]^-$ anion with, for example, $[BMIM]^+$, the viscosity decreases with comparison to $[BMIM][CF_3SO_3]$. The viscosity of $[BMIM]$ $[(CF_3SO_2)_2N]$ is 52 cP, which is 38 cP lower than $[BMIM][CF_3SO_3]$, which has a viscosity of 90 cP, despite it having greater van der Waals interactions.[21] This result is due to the almost complete suppression of hydrogen bonding.

The structure of the cation also plays some role on the viscosity of the ionic liquid. Lower viscosities tend to be a result of low molar mass side chains that have sufficient mobility. Ionic liquids containing the $[EMIM]^+$ cation tend to have the lowest viscosities. Larger or fluorinated alkyl side chains have stronger van der Waals forces and so exhibit a greater viscosity.[21]

An increase in the temperature results in the viscosity of the ionic liquid decreasing. In some cases this can be quite a drastic effect.[42,55] The addition of small amounts of organic cosolvents has also been shown to lower the viscosity significantly.[56-58]

1.4.5 Solubility Characteristics

By carefully selecting the cation and anion of an ionic liquid its solubility properties can be tuned. The effect of the cation is examplified with the solubility of 1-octene in various tosylate melts.[59] It was showed that by increasing the non-polar character of the cation, the solubility of 1-octene increased.

The effect of the anion on solubility characteristics can be seen on the solubility of water in various [BMIM] ionic liquids. [21] Anions such as $[Br]^-$, $[CF_3COO]^-$ or $[CF_3SO_3]^-$ result in very high water solubility. When these anions are replaced by $[PF_6]^-$ or $[(CF_3SO_2)_2N]^-$, a biphasic mixture is formed with water. The water content is very low in either of these ionic liquids. In $[BMIM][(CF_3SO_2)_2N]$ the water content is only 1.4 weight percent at 20 °C.

1.4.6 Solvent Properties

The polarity of a solvent is the most common classification used to describe solvents. Polar solvents are solvents that will dissolve and stabilise dipolar or charged solutes.[1] It is generally thought that ionic liquids are polar solvents because for the most part they fulfil this criteria of polar solvents.[1] Work by Dimroth *et al*. has shown that

using a dye, such as Reichardt's dye, the measurement in the solvatochromic shifts of the absorption band give accurate information on the dipolarity, hydrogen bonding and Lewis acidity of the solvent.[60,61] By definition, water has a value of 1.00 on the E_T^N scale that was introduced by Dimroth *et al* The E_T^N values for a number of alkylammonium nitrate and thiocyanate salts show values in the range of 0.95-1.01.[59] Quaternary ammonium sulfonate salts have been shown to give lower values typically in the range of 0.45-0.65, which is more typical of organic solvents such as DMSO.[60] To separate the dipole-dipole polarizability effects from hydrogen-bonding effects the scales of π* (dipolarity/polarizabiltiy), α (hydrogen-bond-donor acidity) and β (hydrogen-bond-donor basicity) were employed.[60] The greatest variation between ionic liquids was observed in the α and β scales.[63-67] This work has lead to the realisation that ionic liquids can be considered to be polar phases with the solvent properties being largely determined by the ability of the salt to act a hydrogen-bond donor/acceptor.[1, 63-67] The degree of the localisation of the charge on the anion also has a marked effect on its properties. This has a larger effect in the pyridinium and imidazolium based ionic liquids.

1.5 Applications of Ionic Liquids

There has been much work in recent years in the use of ionic liquids for many applications ranging from organic synthesis (Section 1.5.1) to catalysis (Section 1.5.2) to electrochemistry (Section 1.5.3) etc. There have been many discoveries that demonstrate the advantages of using ionic liquids over conventional organic solvents.

1.5.1 Organic Synthesis

1.5.1.1 Diels-Alder Reactions

Water has become a popular solvent for many Diels-Alder reactions, however some reagents are water sensitive and so the possibility of using ionic liquids as a potential polar substitute for water has been investigated. One of the first studies carried out was the reaction of methyl acrylate and cyclopentadiene in $[EtNH_3][NO_3]$.[68]

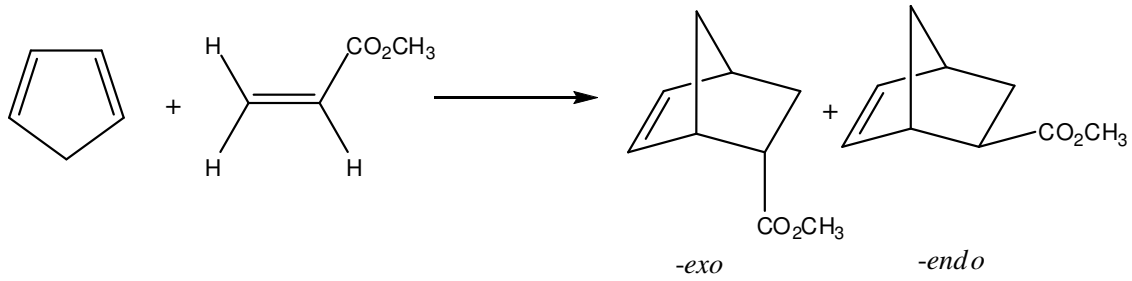


***Figure 1.4**: Diels-Alder formation of both -endo and -exo norborn-5-ene-2-carboxylic acid methyl ester from cyclopentadiene and methyl acrylate.*[1]

This reaction leads to the formation of both *endo* and *exo* products, however the ratio of these products is dependent on the solvent. This phenomenon has been attributed to the 'internal pressure' of the system generated by the solvophobic interactions.[1] This in turn promotes the association of the reagents in a 'solvent cavity' during the activation process. When this reaction is carried out in $[EtNH_3][NO_3]$ there is a very strong preference for the *endo* product and shows enhanced reaction rates over non-polar organic solvents.[68] Other ionic liquids such as $[EMIM][BF_4]$ and $[EMIM][PF_6]$ etc. also show a greater preference for *endo* product and faster reaction rates that non-polar solvents, albeit somewhat weaker. Despite the reaction rates and product selectivity in ionic liquids is better than conventional non-polar organic solvents, water still shows the best rates and product selectivity. Consequently water is still a preferential solvent for many Diels-Alder reactions, however, the use of ionic liquids means that water sensitive reagents can be used for Diels-Alder reactions and the reactions will still occur with relatively fast reaction rates.

1.5.1.2 Friedel-Crafts Reactions

Chloroaluminate ionic liquids have been shown to be very useful for carrying out Friedel-Crafts reactions (Figure 1.5).[19,69] Work by Fremantle has suggested that when carried out in certain ionic liquids, the reaction can reach 100 % conversion in as little as 30 seconds whereas in regular organic solvents it can take around 6-7 hours to achieve only 80 % conversion.[47]

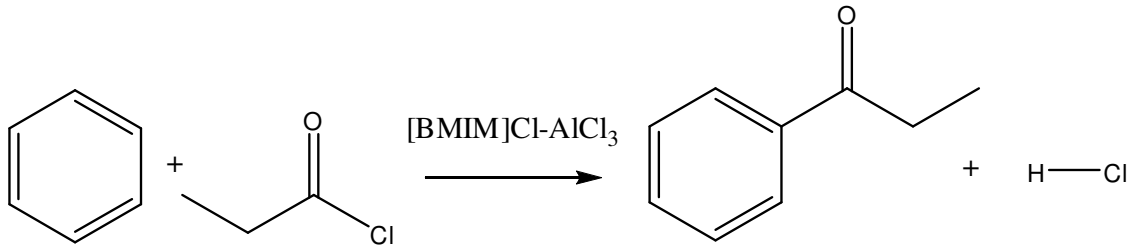


***Figure 1.5**: Friedel-Crafts reaction of benzene and chloroacetone to form propiophenone*

The Lewis acidic $AlCl_3$ catalyses the reaction.[19] However, $AlCl_3$ is very difficult to handle and its recovery from this reaction is difficult so by turning the catalyst into the solvent catalyst so recovery is easier as well as handling.[47,69] This led to the use of chloroaluminate ionic liquids such as [BMIM]Cl-$AlCl_3$. Work in this area has discovered that use of chloroaluminate ionic liquids instead of a conventional organic solvent system used, increases both the yield and reaction kinetics of the reaction. Recovery of the Lewis Acid catalyst simply involves removing the products and excess reagents from the ionic liquid.

1.5.2 Transition Metal Mediated Catalysis

1.5.2.1 Hydrogenation Reactions

Ionic liquids were first used successfully in hydrogenation reaction in 1995 by de Souza[70] and Chauvin[20]. De Souza successfully used a rhodium catalyst to hydrogenate cyclohexene in [BMIM][BF_4]. Whereas Chauvin et.al used [BMIM] ionic liquids using various anions i.e. $[BF_4]^-$, $[PF_6]^-$, $[SbF_6]^-$ to dissolve the 'Osborn complex' [Rh(nbd)(PPh_3)$_2$][PF_6]. This ionic catalyst solution was used for the biphasic hydrogenation of 1-pentene as can be shown in Figure 1.6.

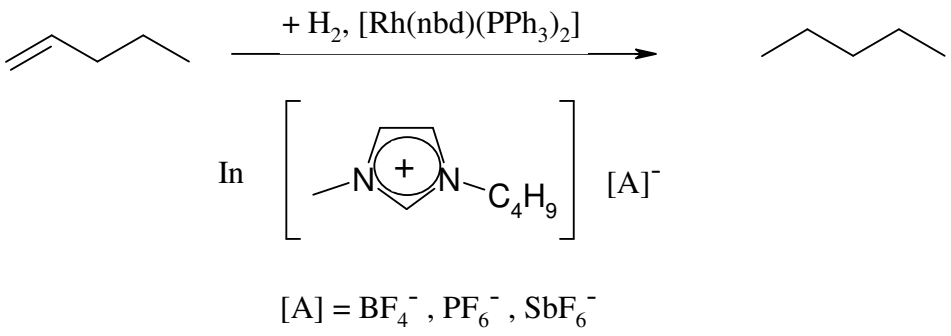


***Figure 1.6**: Hydrogenation reaction carried out in an ionic liquid*

Results for the [BMIM][SbF_6] ionic liquid showed a reaction rate five times greater than the reaction in acetone.[20] The ionic catalysts were ready for reuse after the reaction and could be reused multiple times. In addition, the rate of leaching of the rhodium catalyst into the organic phase was found to be lower than the detectable limit of 0.02%. These reactions showed for the first time that it is possible to 'immobilise' a rhodium catalyst in a polar solution without having to use specially designed ligands.[5,20]

1.5.2.2 Hydroformylation Reactions

Hydroformylation using ionic liquids was reported as early as 1972 by Parshall in which platinum catalysed hydroformylation of ethene in tetraethylammonium trichlorostannate melts.[71] Further work carried out by Wasserscheid *et al*., used the [BMIM]Cl/$SnCl_2$ ionic liquid for the hydroformylation of 1-octene as shown in Figure 1.7.[72]

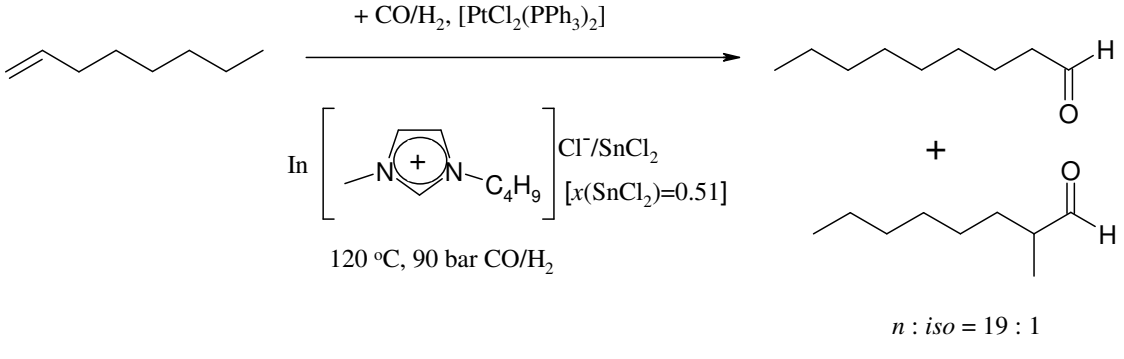


***Figure 1.7**: Hydroformylation reaction carried out in an ionic liquid*[5,72]

Despite the solubility of 1-octene is limited at best in the ionic catalyst phase, the activity of the platinum catalyst is very high (Turnover Frequency (TOF) = 126 h^{-1}).

Product separation is simplified due to the biphasic nature of the system. Parshall and co-worker found no leaching of the catalyst into the organic phase.[72]

1.5.3 Electrochemical Applications

Perhaps the most widely investigated area of ionic liquids is the electrochemical applications. Ionic liquids are good electrical conductors. They can conduct between 10^{-3} and 10^{-2} Ω^{-1} cm^{-1}.[73] The conductivity of conventional organic solvents is negligible since they tend not to contain any species that are capable of carrying charge. Most ionic liquids have been shown to have a electrochemical window of 2-4 V, which is much higher than organic solvents.[73,74] This wider electrochemical window reduces the possibility of any unwanted side reactions occurring. It is these properties that make them interesting for potential use in many electrochemical techniques.

1.5.3.1 Electrodeposition of Metals

Ionic liquids have a much larger electrochemical window than organic solvents. Certain metals can be electrochemically deposited that are otherwise inaccessible to organic solvents.[73,74] The wider electrochemical window reduces the possibility of any side reactions, such as hydrogen evolution occurring that can affect the deposition of the metal onto the substrate,. For example it has been possible to use chloroaluminate ionic liquids to deposit aluminium onto a range of substrates by various electrochemical methods.[75-77] The cluster size of the aluminium deposits on the substrate is affected by the current density, with higher current densities forming smaller crystals.[73] Addition of organic solvents such as dry toluene [78] or benzene [79] have shown that mirror-bright finishes can be obtained after deposition. The miscibility of the choloraluminate based ionic liquids with organic solvents such as toluene or xylene means that the ionic liquid can be washed away from the sample after electrodeposition has taken place. The scope for electrochemically depositing metals successfully is large with the use of ionic liquids. Metals such as In,[80] Sb,[81] Te,[82] Cd,[83] Cu,[84] Ag,[85] Ni,[86] Co,[87] Pd,[87] Au,[89] Zn,[90] Sn,[91] have all successfully been plated onto various substrates.

1.5.3.2 Electrodeposition of Semiconductors

The electrodeposition of semiconductors using ionic liquids is an area still in its infancy but early work has shown promising results.[73] Semiconductors such as GaAs,[92] InSb,[93] ZnTe,[94] Ge,[95] have been effectively electrodeposited using ionic liquids. Because most basic studies of semiconductors have to be carried out under ultrahigh vacuum conditions, ionic liquids are a relatively ideal solution to carry these types of reaction out in. To form certain semiconductors, the kinetic barrier of formation must be overcome. In a vast majority of organic solvents this temperature is too high and the solvent will evaporate. Since ionic liquids can have very high thermal stability, the system can be heated to the temperature required for the semiconductor to form. Temperature has also been shown to affect the quality of the semiconductor deposit.[96]

1.5.3.3 Applications in Electrochemical Devices

Currently, most electrochemical devices contain an aqueous, gel or polymer electrolyte.[74] The conductivity and stability determine the performance and lifetime of the device. Since most room temperature ionic liquids have high conductivities, wide potential windows and stability to moisture and air, they are potential alternatives as unreactive, stable electrolytes for the development of such devices. Electrochemical devices containing a mixture of a BF_4^-/ PF_6^- based ionic liquid and a π-conjugated polymer have been reported and show high stability and lifetime.[97] No appreciable degradation was seen after 1 x 10^6 cycles, which is an improvement by a factor of ten over previous systems. New types of ionic liquid that use DNA as the anion have been developed and are currently being employed in electronic microelectrode arrays for biological purposes.[98] These can be controlled by choice of nucleic acid. The use of room temperature ionic liquids have been explored for the use as electrolytes to improve the conversion efficiency of dye-sensitized solar cells.[99] Double layer capacitors composed of activated carbon electrodes and either [EMIMF][$HF]_{2.3}$, or [EMIM][BF_4] have been reported.[100] The results found that these ionic liquid based systems performance was somewhere between the currently employed aqueous and non-aqueous systems.

1.6 Deep Eutectic Solvents

In Section 1.5 the advantages of using ionic liquids for a wide range of applications has been described, however, there are two key disadvantages of using them. They can be very expensive and difficult to handle. The alternative to using these ionic liquids are Deep Eutectic Solvents (DES) which essentially are an extension of the ionic liquid model.[101-113`] They are basically molten salts that interact by formation of hydrogen bonds or metal halide bonds through the anion of a salt, as opposed to relying purely on electrostatic forces between anion and cation as in the case of ionic liquids. There are various types of DES's which will be explained further on.

The reason they have been termed DES's is because when the two components are added together in the correct ratio a eutectic point can be seen. The eutectic point of a mixture, point *e* in Figure 1.8, is the molar ratio of two compounds which affords the lowest melting point possible. A diagrammatic representation of this can be seen in Figure 1.8.

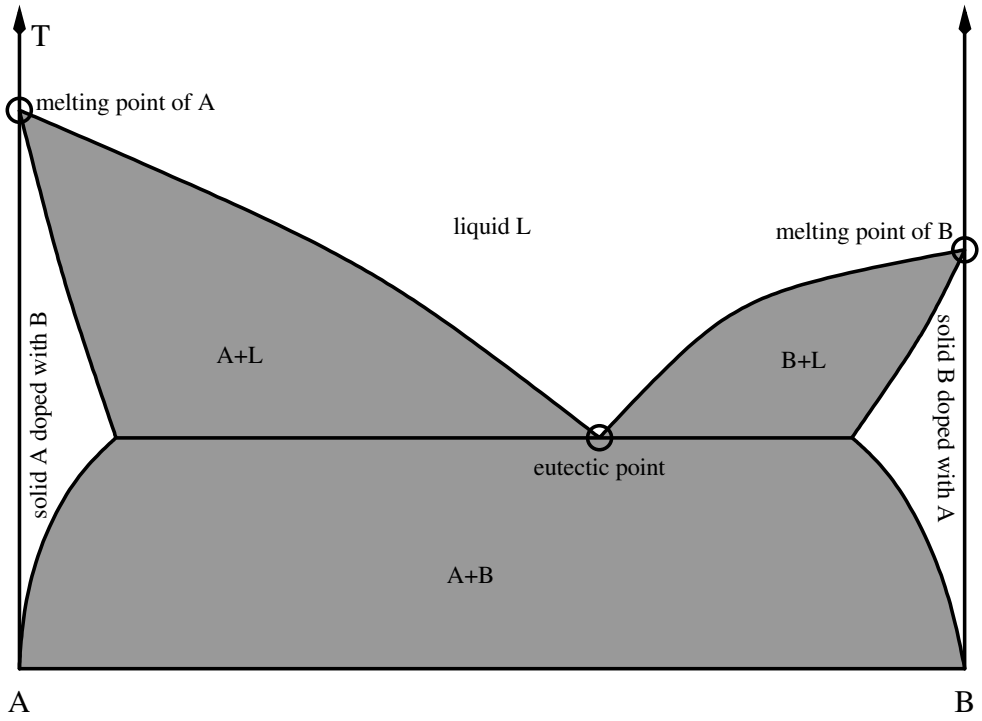


***Figure 1.8**: Diagrammatic representation of the theory behind eutectic mixtures*

The Abbott group has been the first to pioneer this new technology and there has been a large amount of work in the area of eutectic mixtures.[101-113] The work has been varied, and has focused on using metal halide salts, both hydrated and non hydrated, quaternary ammonium salts and hydrogen bond donors. The term DESs, has been coined as a means to differentiate them from true ionic liquids, and also to reflect the large depression in the freezing point of the eutectic mixture. It has been observed that the depression of the freezing point at the eutectic mixture can be as much as several hundred ^{o}C.

1.6.1 Anhydrous Metal Halide Substituted Quaternary Ammonium Salt Mixtures

This type of DES, the form of the complex anion governs the chemistry of these ionic liquids.[101-114] The scope of this type of DES is quite large. Some of the most investigated systems are contain $ZnCl_2$, $SnCl_2$ and $FeCl_3$ all with a variety of quaternary ammonium salts. The metal halides, $SnCl_2$, $ZnCl_2$, and $FeCl_3$, when mixed with choline chloride (ChCl) in a 1:2 ratio, all form materials that have freezing points below 100 °C. This is very in keeping with the idea of DES in that they exhibit very large depressions in freezing point. One of the interesting properties with these liquids is that despite many of them containing metal halides of the same general formula MX_n, there can be quite a difference in the eutectic composition of the system between. For example the eutectic composition of $ZnCl_2$ and ChCl is 2:1, whereas $SnCl_2$ and ChCl is at 2.5:1. This is thought to be due to form of the metal species in the system. Initially at low concentrations of $ZnCl_2$, the amount of $ZnCl_3^-$ relative to the initial $ZnCl_2$ concentration is high. As the molar ratio of $ZnCl_2$ increases past 50 %, the amount of $ZnCl_3^-$ begins to drop in favour of formation of the larger $Zn_2Cl_5^-$ anion. The reason this results in a eutectic composition at 2:1, is because of the nature of these anions. The $ZnCl_3^-$ is a smaller more densely charged anion that has a stronger electrostatic interaction with the chloride ion, thus increasing the lattice energy of the system, resulting in an increase in the freezing point. Hence as the mole fraction reaches 67 %, a reduction in the freezing point is seen since there will be more larger $Zn_2Cl_5^-$ anions than $ZnCl_3^-$. As the mole fraction increases above 67 % an increase in the freezing point is seen again. This is believed to be due to the increase in concentration in the much larger $Zn_3Cl_7^-$ anion,

with the size of the anion being the contributing factor of the increase in freezing point. The reason the $SnCl_2$: ChCl eutectic composition is at 2.5:1 is believed to be due to $SnCl_2$ being a weaker Lewis Acid than $ZnCl_2$ and so it requires a greater amount of $SnCl_2$ to push the equilibrium for the reaction $SnCl_2 + SnCl_3^- \rightleftharpoons Sn_2Cl_5^-$ to the optimum $Sn_2Cl_5^-$ composition. This is good indication that formation of these metal halide based Deep Eutectic Solvents is down to the strength of the interaction between the constituent parts and not simply how many bonds form. From this it can be seen that despite having similar structure metal salts, each system behaves very differently and can obviously tailored for different situations, this what distinguishes them most from the hydrogen bond donor/quaternary ammonium salt systems.

1.6.2 Hydrated Metal Halide/ Substituted Quaternary Ammonium Salt Mixtures

A slight variation on the DESs described in Section 1.6.1 replaces the metal salt with a hydrated metal halide as the anionic complexing species. This general idea has lead to the formation of several useful DES that are liquid at relatively low temperatures.[101-113] For example chromium (in the form of $CrCl_3.6H_2O$), has been used to form a DES with ChCl. It is not possible to form this liquid with the non hydrated version of $CrCl_3$, even if anhydrous $CrCl_3$ is added with 6 molar equivalents of H_2O. This indicates that the coordination sphere of water molecules has a large effect on coordination chemistry around the chromium centre. This would also suggest that this liquid is not just a concentrated aqueous solution and is in fact ionic liquid like. These liquids much simpler to make than conventional metal containing ionic liquid methods. Because they are hydrated metals they are water and air sensitive as well as being much cheaper. The low cost enables their use in large-scale industrial applications. Again these are not truly ionic liquids and hence come under the DES bracket. This is because the system is not made up of one anion and one cationic species that do not fully interact due to the steric bulks of one another. These ionic liquids are slightly different from conventional ionic liquids in that they do not purely interact via ionic interactions but in these cases they interact through the formation of a bond between the anion attached to the quaternary ammonium salt and the metal salt.

1.6.3 Deep Eutectic Solvents Made From Hydrogen Bond Donor and Quaternary Ammonium Salt Mixtures

One of the biggest areas in the DES field is the use of a quaternary ammonium salt and complexing it with a hydrogen bond donor.[105,109] The various hydrogen bond donors that have been looked at involve polycarboxylic acids, polyamides, polyalcohols. These liquids are easy to prepare in a pure state, they are non-reactive with water and many of them are bio-degradable. Despite the basic principle of the way these liquids work being very similar to that of ionic liquids, they are not considered true ionic liquids because they do not contain purely anions and cations. These systems contain a cation and an anion and more importantly there is a hydrogen bond donating species there as well. The end result is similar however, a liquid that is made up a cation and anion is formed, but its route to the reduction of lattice energy of the system is different. This occurs through of the hydrogen bond donor complexing to the anion and withdrawing the electron density of the anion from the cation. This separation results in a weaker anion/cation interaction, which is the way ionic liquids achieve their low melting temperatures.

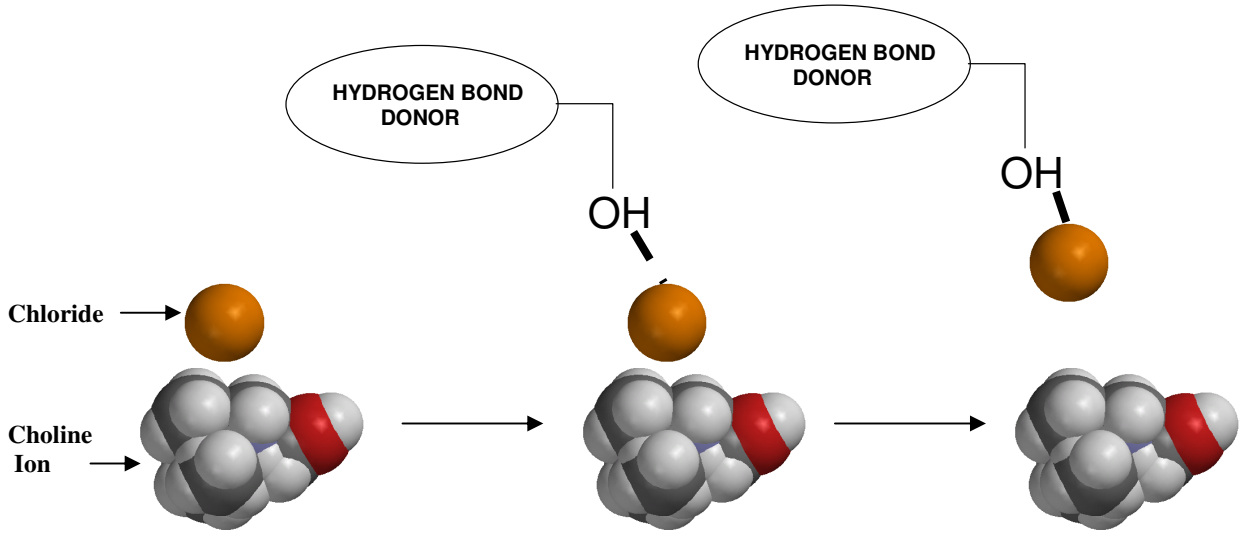


***Figure 1.9**: Diagrammatic representation of how hydrogen bond donors interact with a quarternary ammonium salt*

One of the first DES systems that was investigated, contained ChCl and urea, in a ratio of 1:2 respectively.[105] There has also been a lot of work using carboxylic acids. Abbott *et al.* used phenylic carboxylic acids (phenylpropionic acid and phenylacetic acid) and showed that the eutectic composition with a quaternary ammonium salt was analogous to the urea ChCl case. This 2:1 ratio indicates that it requires two carboxylic acid molecules to complex with the chloride ion to form a eutectic mixture. However it was not possible to form DESs using monocarboxylic acids. By replacing the monocarboxylic acids with dicarboxylic acids with ChCl, there was a eutectic point reached and this eutectic was formed at a ratio of 1:1. This reinforces the theory that each anion is complexed by two hydrogen bonds.[109] It has also been shown that tricarboxylic acids can be used as the hydrogen bond donor. The eutectic point is reached when the composition is between 30-35 molar percentage composition of the acid. Because there are so many hydrogen bond donating groups in these systems, they exhibit the rheology of gels. This is thought that there is extensive bridging of the acids between neighbouring chloride ions. There is also considerable evidence for the depression in the freezing points of the system being in part governed by the molecular weight of the acid, with smaller molecular weight acids exhibiting greater depression in the freezing points. In general, the freezing point of any mixture containing a quaternary ammonium salt and a hydrogen bond donor is dependent on many things but the two most important factors are firstly, how large the individual lattice energies of both the hydrogen bond donor and the quaternary ammonium salt is and secondly the degree to which the anion-hydrogen bond donor interact. If the anion-hydrogen bond donor interaction is strong, the entropy of the system increases, resulting in a more disordered system, therefore offering a lower freezing point. To a certain degree, the depression in the freezing point can be a measure of the entropy change.

These types of DES have been used very successfully for many purposes such as electrochemical applications. Liquids such as Maline 100 (ChCl : malonic acid) and Ethaline 200 (ChCl : 2 x 1,2-ethanediol) have generated a lot of industrial interest in their use for both polishing and plating of metals.

1.6.4 Anhydrous Metal Halide Substituted Hydrogen Bond Donor Mixtures

This is a relatively new area of research in the DESs area.[113] The idea of these systems is to use the metal halide, which in Section 1.6.3 was shown to be the complexing species that disrupts the interaction of the anion/cation interaction of the quaternary ammonium salt, the hydrogen bond donor reverses the role of it. Essentially the metal halide is the replacement for the quaternary ammonium salt. The hydrogen bond donor complexes to one or more of the halides of the metal and withdraws it from the metal centre. The majority of work in this area has been using $ZnCl_2$ as the metal halide and hydrogen bond donors that have been used are acetamide, 1,6-hexanediol, urea and 1,2-ethanediol.[113] Again, as described in Section 1.6.3., the eutectic point is not always as obvious as counting two hydrogen bond donations per halide ion. This is seen with the case for $ZnCl_2$ with both acetamide and 1,2-ethanediol (1:4), but with 1,6-hexandeiol, the eutectic forms at 1:3. This is most probably down to the strength and size of the hydrogen bond donating species, and also the form of the metal halide species.

1.7 The Use of Choline Chloride (hydroxyethyltrimethylammonium chloride) As The Common Quaternary Ammonium Salt

In the area of DESs, the most common quaternary ammonium salt to be used is choline chloride. Initially, this was because of the physical properties it offered when used as part of a DESs. When mixed with most hydrogen bond donors or metal halides, the physical properties were almost always greatly improved over the use of most other quaternary ammonium salts. The depression in freezing point tended to be one of the largest, if not the largest. An example of this is the ChCl: urea mixture. By themselves they have freezing points of 303 °C and 135 °C respectively. It also showed lower viscosities than many other systems containing different quaternary ammonium salts and conductivites were often higher when ChCl was used. The main reason behind ChCl being such a useful quaternary ammonium salt is to do with the fact it is an asymmetric quaternary ammonium salt with a polar functional group, but is also small. The asymmetric nature of this molecule reduces the freezing point of the ionic-

molecular liquid, as does the polar functional group. By combining these two compounds in a ratio of 1:2 (ChCl: urea), the product formed has a freezing point of 12 °C.

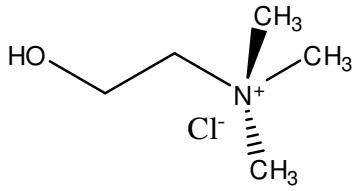


***Figure 1.9**: Structure of Choline Chloride (hydroxyethyltrimethylammonium chloride)*

Perhaps in terms of public perception of this being used as a part of a potential replacement for convential solvents, the fact that choline chloride is essentially non-toxic is a large plus point for its use. Choline chloride is also known as pro-vitamin B4 and has a recommended daily allowance of ~500 mg. It has been used for many years primarily as a chicken feed additive so is very cheap to buy. With the current economic and industrial explosion occurring in China, it has become possible to buy it from Chinese chemical companies for almost £1 per Kg.